

Question 1: Online Shopping System

A database is designed for a large online shopping platform to manage its products, customers, and orders. The platform has many Customers and Products organized by Categories. Customers place Orders containing multiple Products. Each order tracks the quantity and unit price of each product at the time of purchase.

- No product can belong to more than one category.
- A customer may place multiple orders over time.
- The database must contain the following information:
 - o For each customer: customer ID (unique), name, email (unique), phone, and shipping address.
 - o For each product: product ID (unique), name, description, price, and stock quantity.
 - o For each category: category ID (unique), category name, and description.
 - o For each order: order ID (unique), order date, total amount, status, and the customer who placed it.
 - o For each product in an order: quantity ordered and unit price at time of purchase.

ER Modeling Solution – Online Shopping System

Solution 1: Using M:M Relationship (Direct Assignment)

Note: if repetition is not needed for M:M relationship

Entity Types

Customer

- Customer_ID (Primary Key)
- Name
- Email (Unique)
- Phone
- Shipping_Address

Product

- Product_ID (Primary Key)
- Name
- Description
- Price
- Stock_Quantity

Category

- Category_ID (Primary Key)

- Category_Name (Unique)
- Description

Order

- Order_ID (Primary Key)
- Order_Date
- Total_Amount
- Status
- Customer_ID (Foreign Key)

Relationships

Belongs_To

- Between Category and Product
- Cardinality: 1 : M
- One category contains many products.
- Each product belongs to exactly one category.

Places

- Between Customer and Order
- Cardinality: 1 : M
- One customer can place many orders.
- Each order is placed by exactly one customer.

Contains

- Between Order and Product
- Cardinality: M : M
- Relationship Attributes:
 - o Quantity_Ordered
 - o Unit_Price

Solution 2: Resolving M:M Relationship Using Transaction Entity

Note: if repetition is needed for M:M relationship

Entity Types

Customer

- Customer_ID (Primary Key)
- Name
- Email (Unique)
- Phone
- Shipping_Address

Product

- Product_ID (Primary Key)
- Name
- Description
- Price
- Stock_Quantity
- Category_ID (Foreign Key)

Category

- Category_ID (Primary Key)
- Category_Name (Unique)
- Description

Order

- Order_ID (Primary Key)
- Order_Date
- Total_Amount
- Status
- Customer_ID (Foreign Key)

Transaction

- Transaction_Number (Primary Key)
- Quantity_Ordered
- Unit_Price
- Order_ID (Foreign Key)
- Product_ID (Foreign Key)

Relationships

Category_Product

- Between Category and Product
- Cardinality: 1 : M

Customer_Order

- Between Customer and Order
- Cardinality: 1 : M

Order_Transaction

- Between Order and Transaction
- Cardinality: 1 : M
- One order can have many transaction lines (one per product).

Product_Transaction

- Between Product and Transaction

- Cardinality: 1 : M
- One product can appear in many transaction lines over time.

Note: Check if repetition is needed or not for any M:M relationship

Question 2: University Research Database

A database is designed for a university research management system. The university has multiple Faculties, each containing several Researchers. Researchers work on Research Projects and publish Papers. Each paper may have multiple researchers as co-authors. A researcher can be the lead author of many papers, and a paper has exactly one lead author. Each research project is funded by one or more Funding Agencies.

- Each researcher belongs to exactly one faculty.
- A researcher can work on multiple projects simultaneously.
- Each project can be funded by multiple agencies, and each agency can fund multiple projects, recording the funded amount and start year.
- The database must contain the following information:
 - o For each faculty: faculty ID (unique), name, dean name, and location.
 - o For each researcher: researcher ID (unique), name, specialization, email, and rank.
 - o For each paper: paper ID (unique), title, publication date, journal name, and lead author.
 - o For each project: project ID (unique), title, start date, end date, and budget.
 - o For each funding agency: agency ID (unique), name, and country.

ER Modeling Solution – University Research Database

Solution 1: Using M:M Relationship (Direct Modeling)

Note: if repetition is not needed for M:M relationship

Entity Types

Faculty

- Faculty_ID (Primary Key)
- Name
- Dean_Name
- Location

Researcher

- Researcher_ID (Primary Key)
- Name

- Specialization
- Email (Unique)
- Rank
- Faculty_ID (Foreign Key)

Paper

- Paper_ID (Primary Key)
- Title
- Publication_Date
- Journal_Name
- Lead_Author_ID (Foreign Key → Researcher)

Project

- Project_ID (Primary Key)
- Title
- Start_Date
- End_Date
- Budget

Funding_Agency

- Agency_ID (Primary Key)
- Name
- Country

Relationships

Belongs_To

- Between Faculty and Researcher
- Cardinality: 1 : M
- One faculty has many researchers.
- Each researcher belongs to exactly one faculty.

Leads

- Between Researcher and Paper
- Cardinality: 1 : M
- One researcher can be lead author of many papers.
- Each paper has exactly one lead author.

Co_Authors

- Between Researcher and Paper
- Cardinality: M : M
- A paper can have many co-authors.
- A researcher can co-author many papers.

Works_On

- Between Researcher and Project
- Cardinality: M : M
- A researcher can work on many projects.
- A project can have many researchers.

Funds

- Between Funding_Agency and Project
- Cardinality: M : M
- Relationship Attributes:
 - Funded_Amount
 - Start_Year

Solution 2: Resolving M:M Relationships Using Transaction Entity

Note: if repetition is needed for M:M relationship

Entity Types

Faculty

- Faculty_ID (Primary Key)
- Name
- Dean_Name
- Location

Researcher

- Researcher_ID (Primary Key)
- Name
- Specialization
- Email (Unique)
- Rank
- Faculty_ID (Foreign Key)

Paper

- Paper_ID (Primary Key)
- Title
- Publication_Date
- Journal_Name
- Lead_Author_ID (Foreign Key → Researcher)

Project

- Project_ID (Primary Key)
- Title
- Start_Date

- End_Date
- Budget

Funding_Agency

- Agency_ID (Primary Key)
- Name
- Country

Transaction1 (Co-Authorship)

- Transaction_Number1 (Primary Key)
- Researcher_ID (Foreign Key)
- Paper_ID (Foreign Key)

Transaction2 (Project Assignment)

- Transaction_Number2 (Primary Key)
- Researcher_ID (Foreign Key)
- Project_ID (Foreign Key)

Transaction3 (Funding)

- Transaction_Number3 (Primary Key)
- Agency_ID (Foreign Key)
- Project_ID (Foreign Key)
- Funded_Amount
- Start_Year

Relationships

Faculty_Researcher

- Between Faculty and Researcher
- Cardinality: 1 : M

Researcher_Lead

- Between Researcher and Paper (lead authorship)
- Cardinality: 1 : M

Researcher_Transaction1

- Between Researcher and Transaction1
- Cardinality: 1 : M
- One researcher can co-author many papers.

Paper_Transaction1

- Between Paper and Transaction1
- Cardinality: 1 : M
- One paper can have many co-authors via transactions.

Researcher_Transaction2

- Between Researcher and Transaction2
- Cardinality: 1 : M

Project_Transaction2

- Between Project and Transaction2
- Cardinality: 1 : M

Agency_Transaction3

- Between Funding_Agency and Transaction3
- Cardinality: 1 : M

Project_Transaction3

- Between Project and Transaction3
- Cardinality: 1 : M

Note: Check if repetition is needed or not for any M:M relationship

Question 3: Healthcare Clinic Database

A database is designed for a multi-branch healthcare clinic to manage its doctors, patients, appointments, and medical records. The clinic has multiple Branches. Each branch employs several Doctors, each specializing in one or more medical fields. Patients register at the clinic and can visit any branch. During each visit, a doctor examines the patient and creates a Medical Record.

- Each doctor is employed by exactly one branch.
- A doctor can have one or more specializations.
- A patient can have appointments with multiple doctors across different branches.
- Each appointment involves exactly one doctor and one patient, and produces one medical record.
- Medicines can be prescribed in medical records; each prescription records the dosage and duration.
- The database must contain the following information:
 - o For each branch: branch ID (unique), location, phone number, and manager name.
 - o For each doctor: doctor ID (unique), name, email, years of experience, and specialization(s).
 - o For each patient: patient ID (unique), name, date of birth, blood type, and contact details.
 - o For each appointment: appointment ID (unique), date, time, and status.
 - o For each medical record: record ID (unique), diagnosis, notes, and appointment date.
 - o For each medicine: medicine ID (unique), name, type, and manufacturer.

ER Modeling Solution – Healthcare Clinic Database

Solution 1: Using M:M Relationship (Direct Modeling)

Note: if repetition is not needed for M:M relationship

Entity Types

Branch

- Branch_ID (Primary Key)
- Location
- Phone_Number
- Manager_Name

Doctor

- Doctor_ID (Primary Key)
- Name
- Email (Unique)
- Years_of_Experience
- Specialization (Multivalued Attribute)
- Branch_ID (Foreign Key)

Patient

- Patient_ID (Primary Key)
- Name
- Date_of_Birth
- Blood_Type
- Contact_Details

Appointment

- Appointment_ID (Primary Key)
- Appointment_Date
- Appointment_Time
- Status
- Doctor_ID (Foreign Key)
- Patient_ID (Foreign Key)

Medical_Record

- Record_ID (Primary Key)
- Diagnosis
- Notes
- Record_Date
- Appointment_ID (Foreign Key)

Medicine

- Medicine_ID (Primary Key)
- Name
- Type
- Manufacturer

Relationships

Employs

- Between Branch and Doctor
- Cardinality: 1 : M
- One branch employs many doctors.
- Each doctor works at exactly one branch.

Books

- Between Patient and Appointment
- Cardinality: 1 : M
- One patient can have many appointments.
- Each appointment is booked by exactly one patient.

Conducts

- Between Doctor and Appointment
- Cardinality: 1 : M
- One doctor can conduct many appointments.
- Each appointment is conducted by exactly one doctor.

Generates

- Between Appointment and Medical_Record
- Cardinality: 1 : 1
- Each appointment generates exactly one medical record.

Prescribed_In

- Between Medical_Record and Medicine
- Cardinality: M : M
- A medical record can include many prescribed medicines.
- A medicine can appear in many medical records.
- Relationship Attributes:
 - o Dosage
 - o Duration

Solution 2: Resolving M:M Relationship Using Transaction Entity

Note: if repetition is needed for M:M relationship

Entity Types

Branch

- Branch_ID (Primary Key)
- Location
- Phone_Number
- Manager_Name

Doctor

- Doctor_ID (Primary Key)
- Name
- Email (Unique)
- Years_of_Experience
- Specialization (Multivalued Attribute)
- Branch_ID (Foreign Key)

Patient

- Patient_ID (Primary Key)
- Name
- Date_of_Birth
- Blood_Type
- Contact_Details

Appointment

- Appointment_ID (Primary Key)
- Appointment_Date
- Appointment_Time
- Status
- Doctor_ID (Foreign Key)
- Patient_ID (Foreign Key)

Medical_Record

- Record_ID (Primary Key)
- Diagnosis
- Notes
- Record_Date
- Appointment_ID (Foreign Key)

Medicine

- Medicine_ID (Primary Key)
- Name
- Type
- Manufacturer

Transaction (Prescription)

- Transaction_Number (Primary Key)
- Dosage
- Duration
- Record_ID (Foreign Key)
- Medicine_ID (Foreign Key)

Relationships**Branch_Doctor**

- Between Branch and Doctor
- Cardinality: 1 : M

Patient_Appointment

- Between Patient and Appointment
- Cardinality: 1 : M

Doctor_Appointment

- Between Doctor and Appointment
- Cardinality: 1 : M

Appointment_Record

- Between Appointment and Medical_Record
- Cardinality: 1 : 1

Record_Transaction

- Between Medical_Record and Transaction
- Cardinality: 1 : M
- One medical record can have many prescription transactions.

Medicine_Transaction

- Between Medicine and Transaction
- Cardinality: 1 : M
- One medicine can appear in many prescription transactions.

Note: Check if repetition is needed or not for any M:M relationship